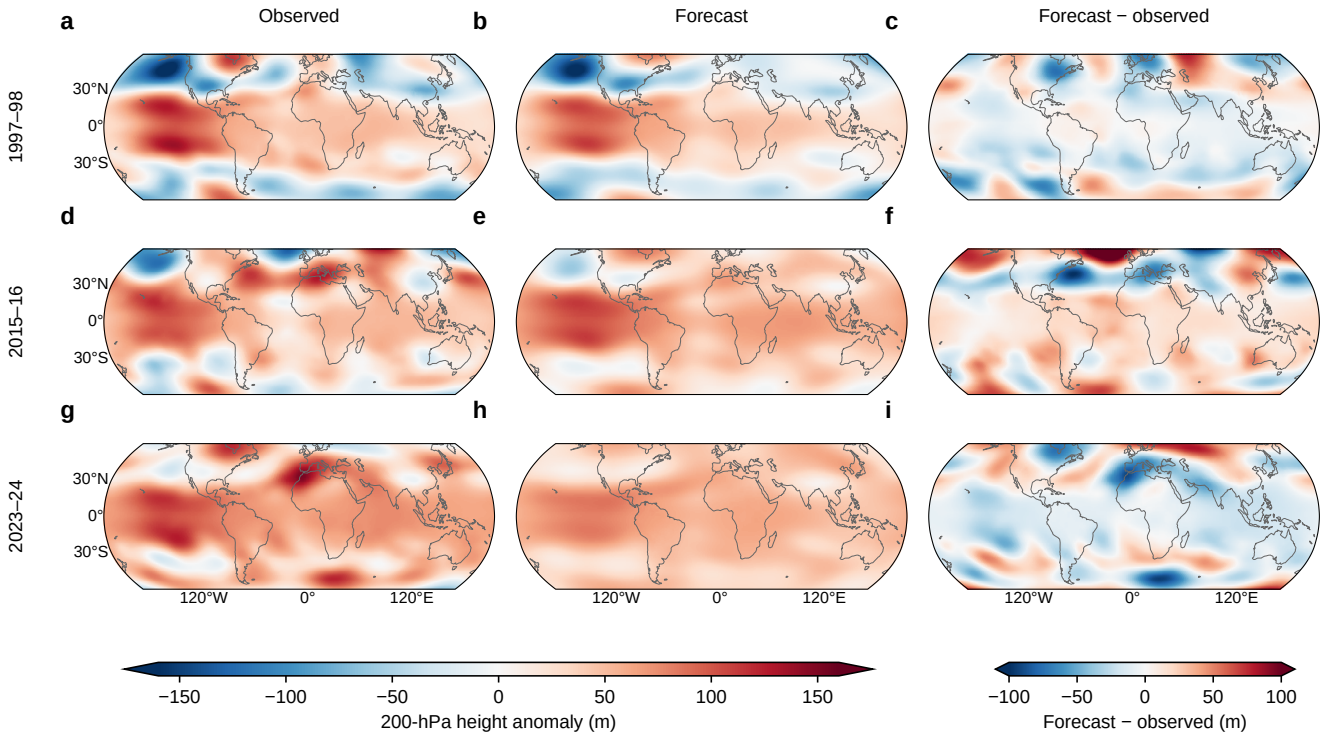
Supplementary Fig. 6 | Observed and forecast upper-level circulation in three El Niño events



Rows show December–February (DJF) of 1997–1998, 2015–2016 and 2023–2024; columns show ERA5 200-hPa geopotential-height anomalies, multimodel forecast anomalies and forecast minus observed anomalies. Forecasts first average members within each model, then equally weight the three available models and the eight common future leads. Observed and forecast fields share the ±160 m colour scale; differences share ±100 m, with extensions marking values beyond the limits. Red denotes positive height anomalies or positive forecast error, blue negative values. Anomalies use calendar-month means from 1992–2020; February 2018 is omitted consistently from all model and observation climatologies because the corresponding GEOS-S2S field is missing. No event DJF month is excluded. Lead 9 is absent from the fixed three-model cohort. Maps retain their native 1° resolution without added smoothing.

Models are COLA–RSMAS–CCSM4, NASA–GEOS-S2S and GFDL–SPEAR. These are seasonal composites, not isolated causal ENSO responses. Signed averaging can cancel opposing errors; Fig. 8 also evaluates RMSE.